御 Mitsue Project
御杖プロジェクト

NAIST SEMINAR · 13 JANUARY 2027

Planted for a market that *no longer exists.*

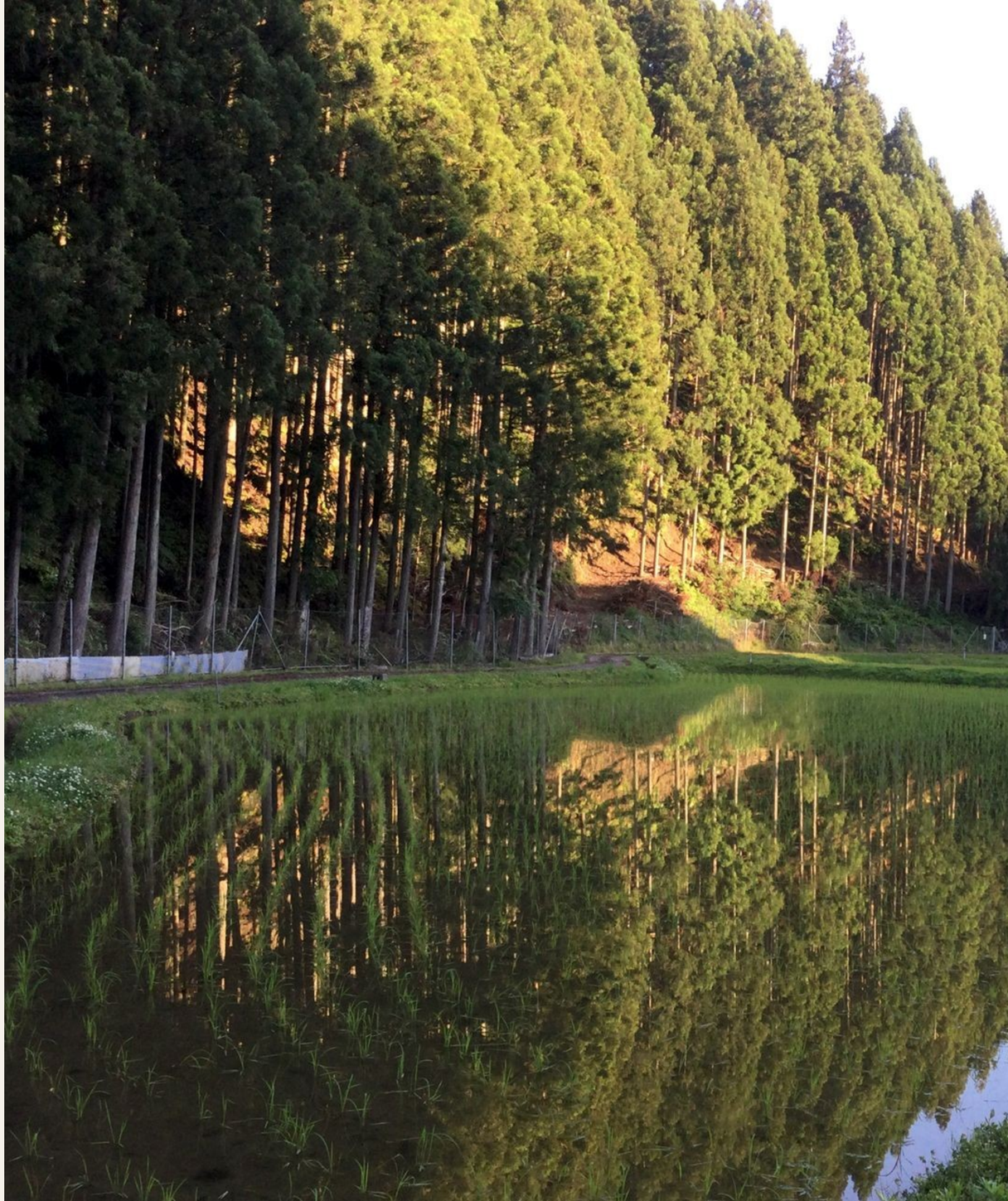
いまは無い市場のために植えられた森

A forest, a village, and a sideways idea: history, today, and what we are still working out.

01–10 JAMES · HOW WE GOT HERE

11–20 ROB · WHERE WE ARE, WHAT WE TRY

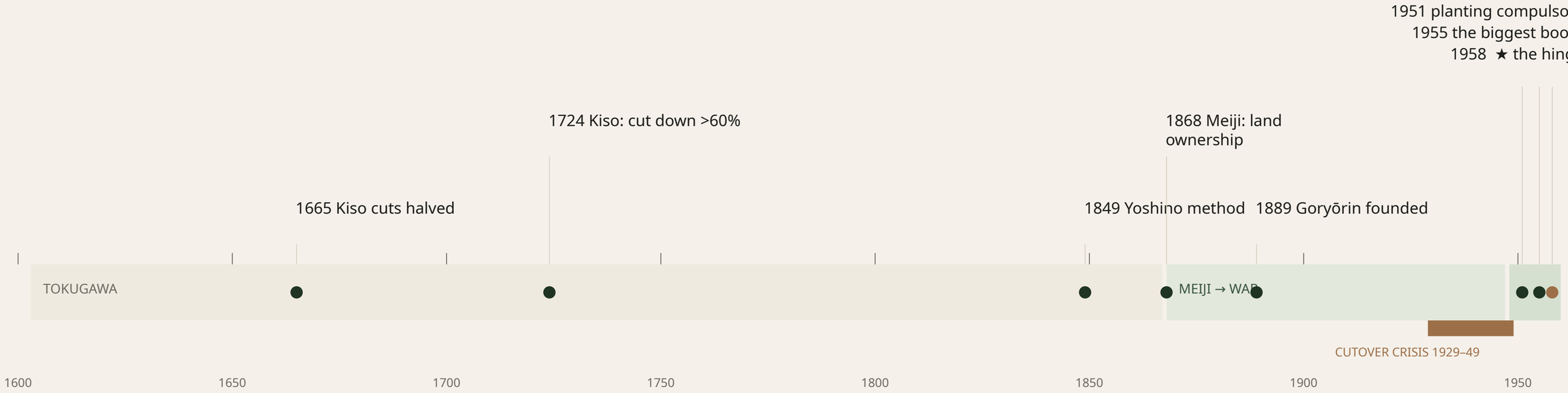
JAMES HILL OPENS · ROB OUDENDIJK CONTINUES



HISTORY 02 / 10 · 1603–1958 · THE WHOLE STORY IN ONE STRIP

JAMES · TO EXPAND

Three centuries, *one strip.*



The punchline arrives in 1958: the trees we see today were planted just as the demand behind them began to fade.

→ ROB · 11

The same trees, today: what it looks like on the ground.

HISTORY 03 / 10 · 1603–1849 · TENURE AND THE YOSHINO METHOD

JAMES · TO EXPAND

Who owns a tree on shared land? *And how do you grow one?*

TENURE

Land is feudal or communal; private ownership is banned in principle. Lords' forests in Kiso are guarded and theft is severely punished.

WAVE 1 · LATE 1600s

Edo and Osaka boom: a 50-year timber investment pays for the first time. Custom starts to recognise private ownership of planted trees on communal land (Yoshino, Owase).

CUTS AND COUNTS

Kiso cuts halved in 1665, down by more than 60% in 1724 and held for 30 years. First forest inventory 1779. Edo reaches 1,000,000 people in 1801.



Okino Takao, Kurobane domain, 1849: a citable Edo-era schedule that still looks familiar. Also: 1750s Yamaguni guild licence class for plantation growers.

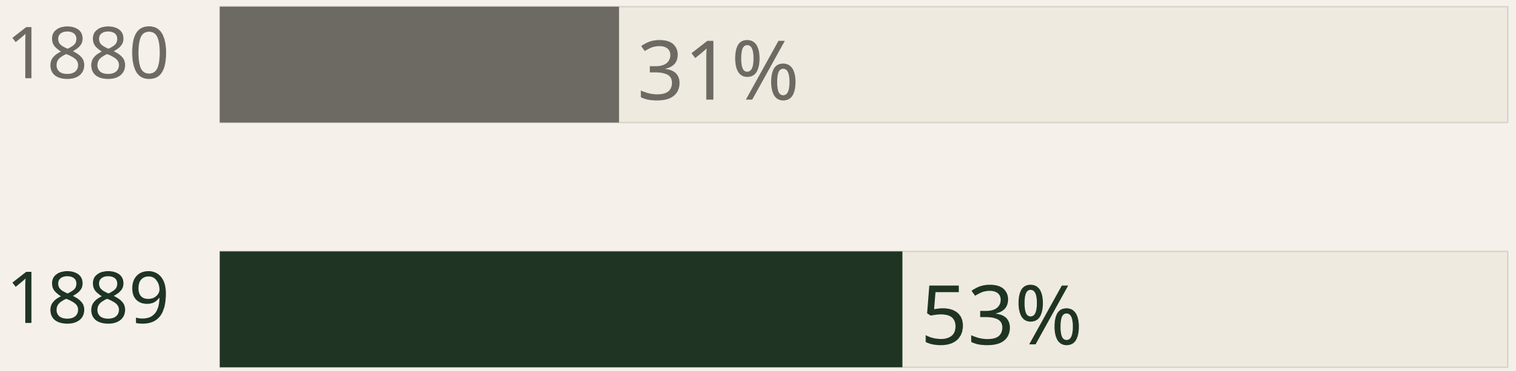
→ ROB · 13

Thin early, thin again. Who does the thinning today?

HISTORY 04 / 10 · 1868–1889 · KAN / MIN

JAMES · TO EXPAND

Government or private? *Six policy reversals in five years.*



share of Japan's woodland held by the government

1868 formal ownership introduced; national forests carved from communal and former feudal land.
1871–76 the state tries to sort all woodland into kan or min.

官有地 kan'yuchi

government. the state wants sanrin: dense conifer

民有地 min'yuchi

private. villagers need gen'ya: scrub for fuel and fodder

共有地 koyuchi

shared use. a third category, because the first two did not fit real life

→ ROB · 11

Over 90% of Mitsue's forest is private: the village is the 'min' side of this story.

HISTORY 05 / 10 · 1889–1947 · GORYŌRIN

JAMES · TO EXPAND

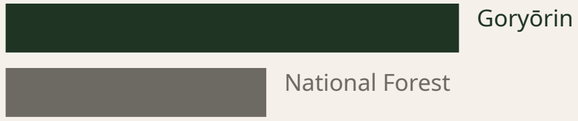
For 58 years, forestry was a *real business.*

1.6 M ha
carved out in 1889–90 as the Imperial Household's own income-producing forest estate

40 %
net margin on gross income at the 1936–45 peak, against 23% for the much larger National Forest

31 %
of Imperial Household income in 1940 (19% in 1916)

70 %
of Household wealth by 1946: ¥2.6 bn of cumulative asset value



1889 founded

1892 turns profitable

1942 Hundred Year Plan suspended, 'special harvesting'

1947 dissolved

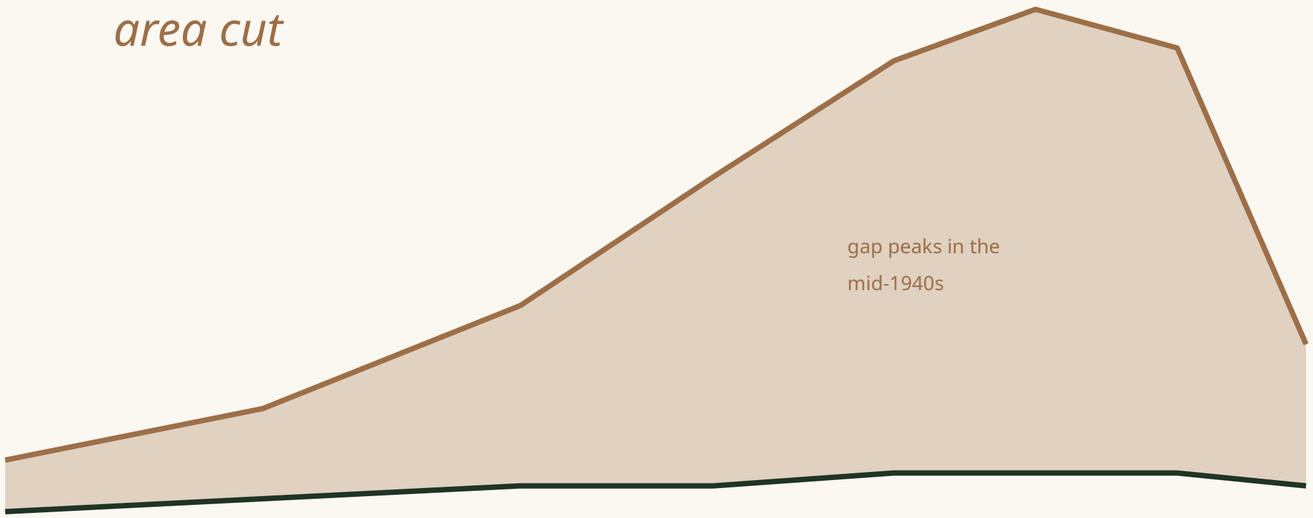
→ ROB · 14

A buyer that pays: the idea we are trying to rebuild with a power plant.

HISTORY 06 / 10 · 1929–1949 · THE CUTOVER CRISIS

JAMES · TO EXPAND

Twenty years in a row, more was cut *than planted.*



1.49 M ha

cutover land in 1948, about 6% of Japan's forest area, concentrated near villages

8.6 → 10.97

million koku harvested, 1942 to 1944: paid for by cutting reforestation and aftercare

1948 forecast: wood exhausted in 15–33 years. Wrong, but it drove everything after.

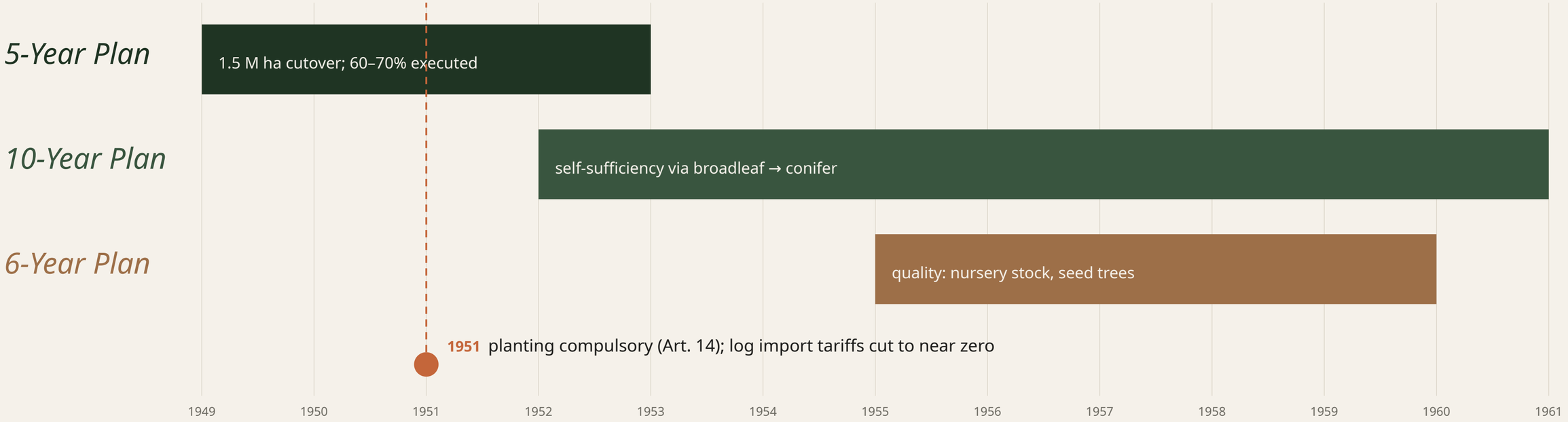
→ ROB · 12

Cut > planted: when short-term demand beats the plan. We call it 'no pull to replant'.

HISTORY 07 / 10 · 1949–1961 · POSTWAR PLANS

JAMES · TO EXPAND

Three plans in twelve years, *and planting becomes the law.*



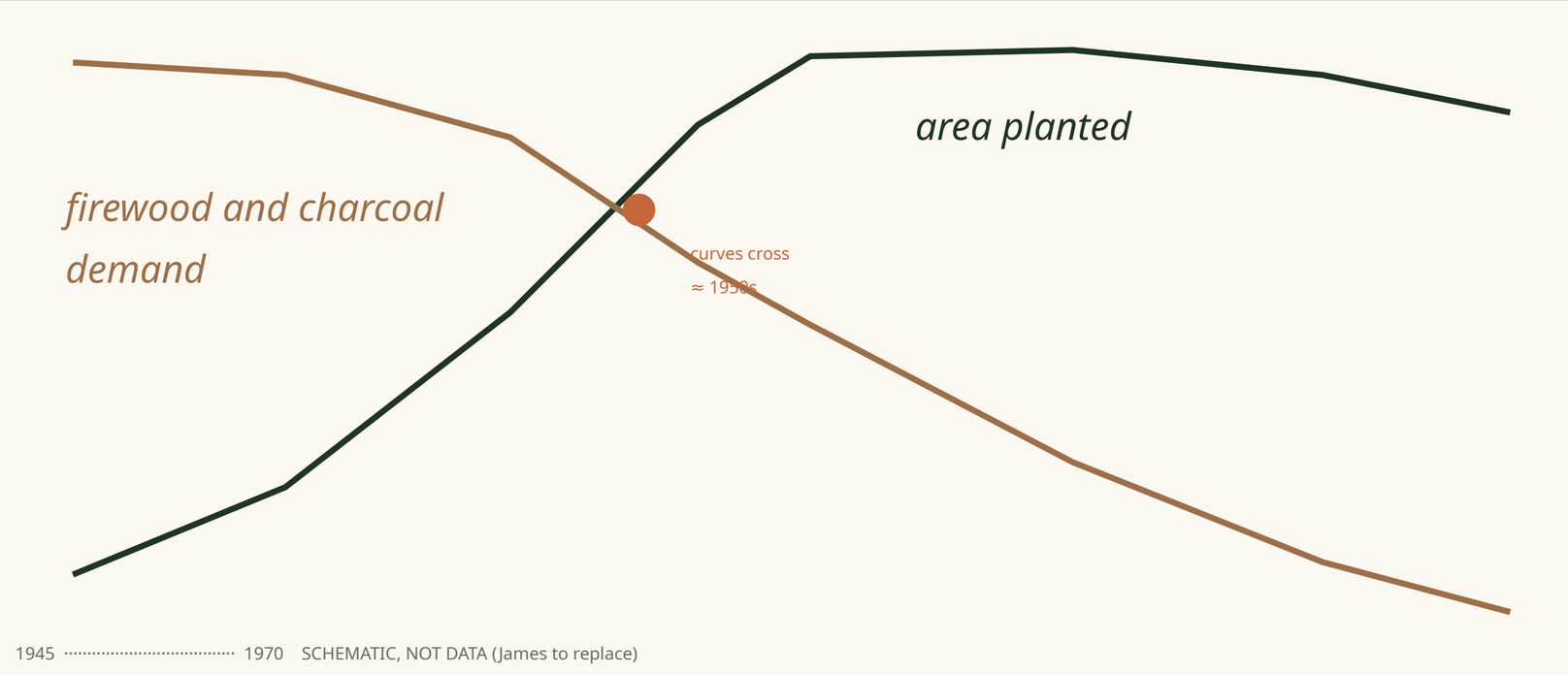
→ ROB · 15

Broadleaf was turned into conifer. We now want to put broadleaf back, where it makes sense.

HISTORY 08 / 10 · 1950–1955 · TWO CURVES CROSS

JAMES · TO EXPAND

Planting peaks. *The reason for planting fades.*



1950

The demand floor drops out

Fossil fuels replace firewood and charcoal; chemical fertiliser replaces green fertiliser; machines replace draft animals.

1954–55

Peak planting

Highest annual area under the 10-Year Plan in 1954. Wave 3, the biggest boom, runs 1955–1970.

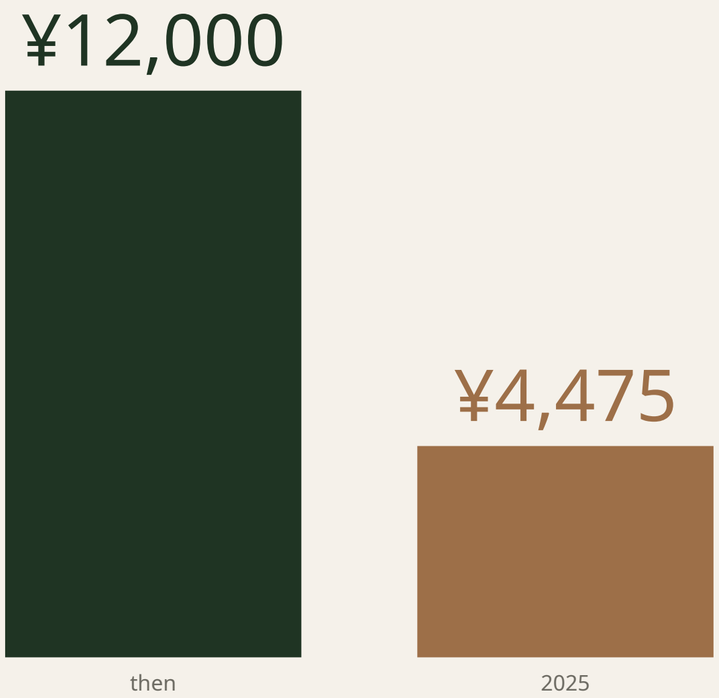
→ ROB · 14

Demand for the trees fell away. Can a small power plant bring some of it back?

HISTORY 09 / 10 · 1951–2025 · SEEDS AND THE CRASH

JAMES · TO EXPAND

Planted as a retirement fund, *then the market crashed.*



A-grade sugi per m³ (years and series to be sourced)

1951

Import tariffs to near zero

Sensible at the time; it set up self-sufficiency to fall for decades.

1956

Forest Development Corporation

Long-term debt plants the least attractive land against a harvest 40–50 years out. After 1980 all 47 prefectural corporations fall into debt distress.

1970s–80s

Cheap imports

The market crashes before the planted trees are old enough to cash in. Only A-grade sells today.

→ ROB · 15

Only A-grade sells today. What do we do with B, C and D?

JAMES · TO EXPAND

1958

The trees are older than *the demand that justified planting them.*



We are all born without clothes.
We are all the same. Humans.

ROB, NOVEMBER 1958

Deep inside the biggest boom (1955–70), planting compulsory since 1951, self-sufficiency still about 95%. The sugi and hinoki in Mitsue's mountains were planted in exactly this window. Over to Rob: today.

→ ROB · 11

James hands over: what is standing on those slopes now?



ROB 01 / 10 · TODAY · THE SETTING

A village of 1,400 people, *a forest of 7,000 hectares.*

≈ 1,400

residents

≈ 93 %

forest cover

≈ 7,050 ha

forest, over 90% private

50–60 yr

age of the main sugi and hinoki stands

Many owners are elderly or live elsewhere. At today's prices a cut sugi barely pays for the cutting. Still checking: how many owners, how many parcels, who inherits.

← JAMES · 10

The sugi and hinoki planted in exactly that window.

ROB 02 / 10 · THE PROBLEM

Three problems with one shape: *nobody is paid to fix it.*



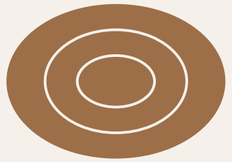
Not enough hands

The cooperative is small and has few young people. Harvest at scale needs machines and trained operators. Thinning is planned, but most of it is cut and left.



A ticking clock

A large share of the forest reaches harvest age inside the same ~25-year window in which new small-scale power technology matures. Nice timing, if we use it.



No pull to replant

Cutting sugi for cheap lumber does not pay enough to put anything back. Short-term profit, repeat forever. Nobody has a reason to break the cycle.

← JAMES · 7 — *A ticking clock: the 1950s planting wave all reaches harvest age together.*

← JAMES · 9 — *No pull to replant: the market crashed before the trees grew up.*

ROB 03 / 10 · THE PROBLEM IN NUMBERS

STILL RESEARCHING

Most of the thinning wood *stays in the forest.*

18,140 m³ cut and left in the forest

2,574 m³ taken out

PLANNED THINNING FY2024–26 · 113 PARCELS · ≈ 222 ha · 4 DISTRICTS



≈ 88 %

stays in the forest: about 860 m³ a year actually comes out

That cut-and-left wood is a fuel resource, if it can be collected at a price that makes sense. Steep slopes and missing roads are why it is left.

Still checking: how much is reachable with a winch-assisted machine, and what an earlier '120–300 m³/yr' cooperative figure really measured.

← JAMES · 3

Thin early, thin again: the rule is 170 years old. Here nobody is there to do it.

Sources: Village forestry plan R6–R8 (FY2024–26); village hall (Oba)

ROB 04 / 10 · THE IDEA

Pull, don't push: *give the wood a buyer that pays.*

1 *Thin and clear*

Cut-to-length machines on steep sugi

2 *Sort A / B / C / D*

A and B to the sawmill, C and D to chips

3 *Chip and dry*

Local chipping, drying with CHP heat

4 *Small CHP*

Gasification plant: power and heat, 24 h









Why 'pull'. Waiting for a subsidy pushes wood out once, then stops. A buyer that wants fuel 24 hours a day pulls wood out every year, and can pay for the replanting.

Then replant. Cleared stands go back to native broadleaf (oak, chestnut, konara) for wildlife and watershed, with fast-growing kiri on part of the land. Species chosen with the NGO more trees.

← JAMES · 5

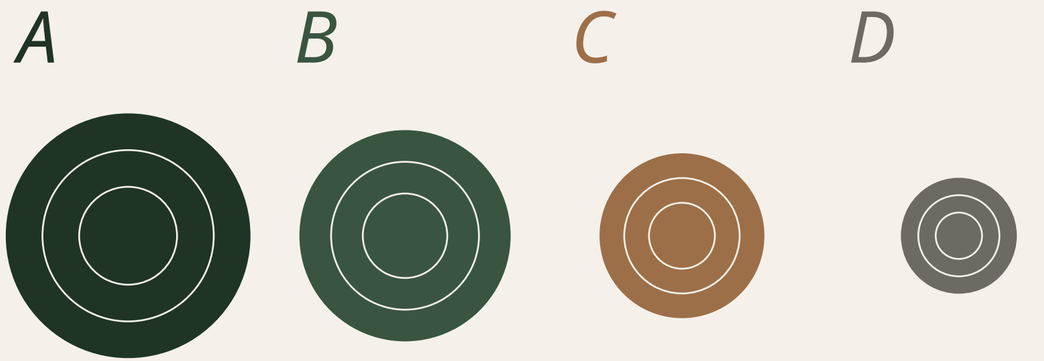
Goryōrin: the last time forestry was a business. Someone needed the wood and paid for it.

ROB 05 / 10 · WHAT A TREE IS WORTH, AND WHAT COMES NEXT

STILL RESEARCHING

Sell the straight logs, *burn the rest, replant the hill.*

WHAT A TREE IS WORTH



Straight, clean: sawlog

Some knots: sawlog

Crooked, small: chips

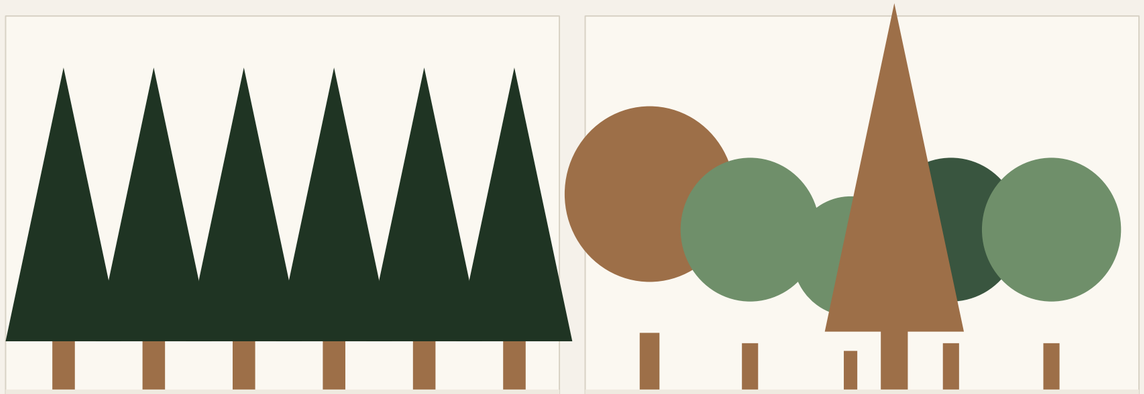
Tops, bark: stays on slope

Only A-grade sells today. With a CHP plant as buyer, C and D wood suddenly has a customer too.

← JAMES · 9

Only A-grade sells: the crash.

THEN REPLANT



pollen, landslide risk

acorns for wildlife

Native broadleaf (oak, chestnut, konara), chosen with the NGO more trees, plus kiri (paulownia) as a fast fuel crop on part of the land. Mitsue growth data requested from SANSEI. Still checking.

← JAMES · 7

The 1952 plan turned broadleaf into conifer. We go the other way.

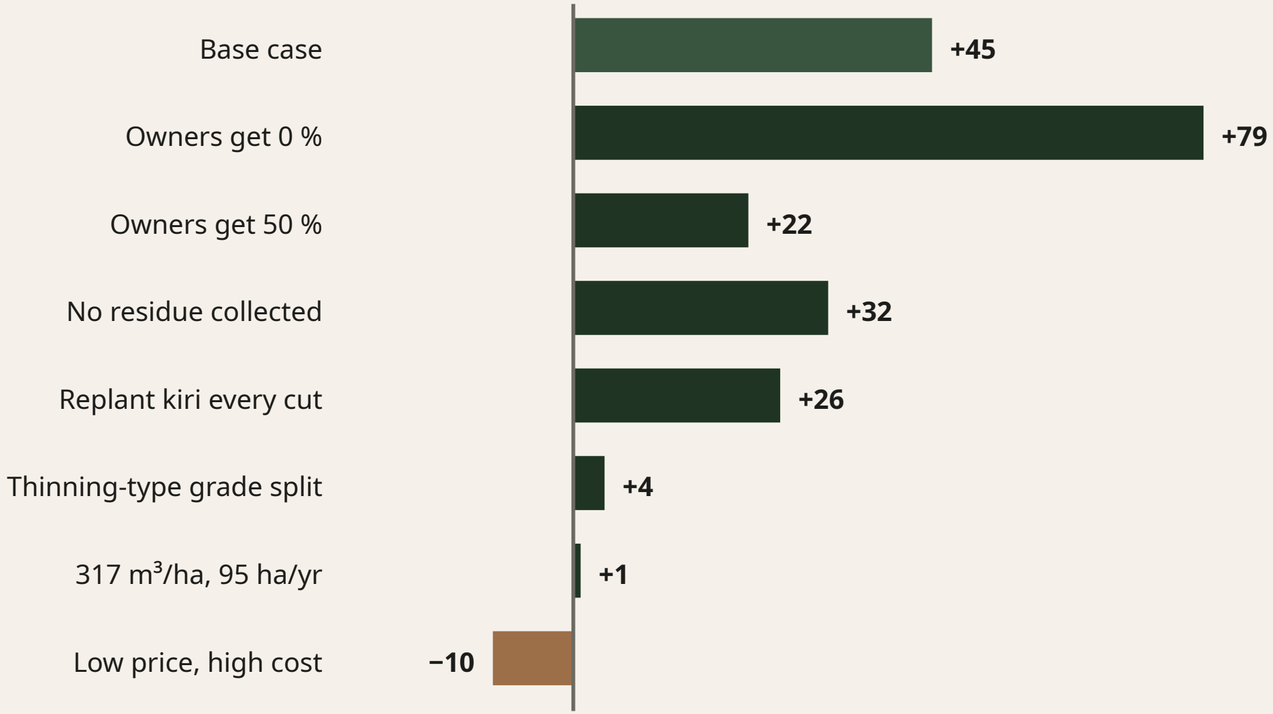
Sources: Grade descriptions simplified; model v1.2 (61% A national average, 22% A in thinning-type stands)

ROB 06 / 10 · THE FIRST NUMBERS

PLACEHOLDER PRICES

Our first model says plus, *then shows how fragile that is.*

STEADY-STATE SYSTEM NET, ¥M PER YEAR



Not a forecast.

Base: 50 ha/yr clear-cut at 600 m³/ha, sawlogs sold, C+D chipped, half replanted to kiri, owners paid 30%.

Four unknowns decide it: wood per hectare (600 assumed, village plan 277–317), A-grade share (61% national, 22% thinning-type), landowner share (30%), residue price (¥4,000/t placeholder).

Left out: carbon credits, data-center income, machines, wages.

← JAMES · 6

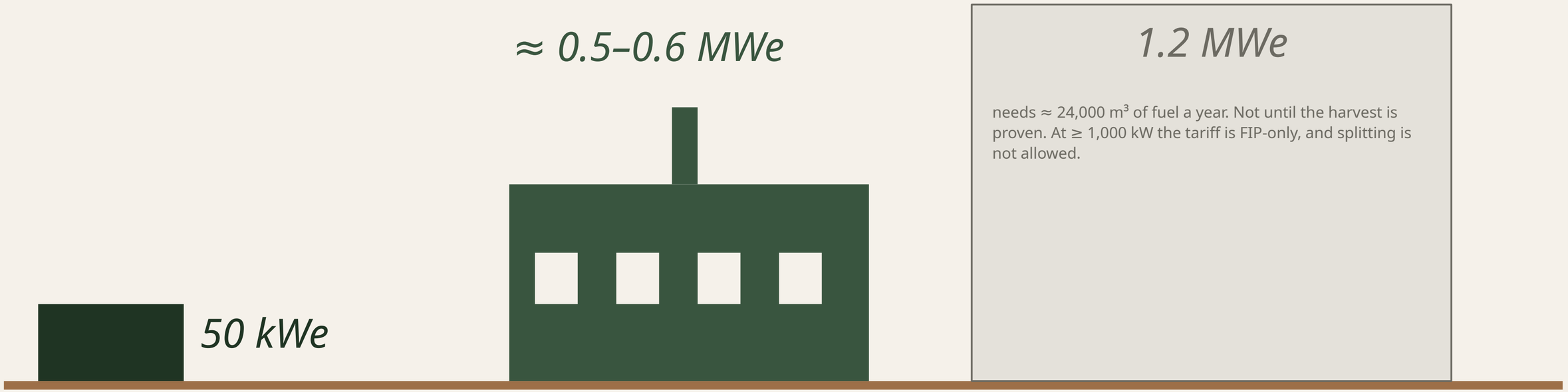
Wartime override: what happens when short-term demand beats the plan. Our model has the same weak spot.

Sources: kiri_sugi_integrated_system_model_2026-11-03.xlsx v1.2. All prices and costs are placeholders, not quotes.

ROB 07 / 10 · THE ENERGY SIDE

STILL RESEARCHING

Start small to learn. *Size the real plant to the wood we can get.*



Prototype in a 20 ft container. ≈ 450–600 t dry wood a year. A learning step: efficiency only ~12–20% at this size.

First commercial size. 23–28% efficiency for gasifier plus engine at this scale. Fuel mostly C+D wood and residue.

NOT TO SCALE

Also: Japanese maker only; gasification + engine beats steam below ~1 MWe; grid at 6.6 kV has about 3 MW locally, none upstream; fuel may be bought from neighbouring villages.

Sources: Energy, finance and resources overview v1.1 (2026-11-03)

ROB 08 / 10 · THE HOMEWORK, AND WHAT WE STILL DO NOT KNOW

STILL RESEARCHING

We read, we asked, we counted. *Then we listed what we do not know.*

45
acts on forestry, biomass, 協力隊 , siting read and summarised

113
thinning parcels read one by one

2
rounds of questions to Komatsu on steep-slope harvesting

v1.2
of the spreadsheet model: clear-cut, logs, chips, kiri, CHP, owners

5+
forest, village hall, Uda City and Tenkawa visits

100%
open in a public repository, to check or challenge

OPEN QUESTION	WHO	STATUS
Stand volume, stumpage, who gets the sale income	**Village hall (Oba)**	asked 6 Nov
Slope limits with and without winch, road width, yearly output	**Komatsu**	follow-up sent
Real CHP efficiency and cost at 0.6–0.8 MWe	**Maker quotes**	to request
Kiri growth on Mitsue slopes	**SANSEI planting data**	asked 2 Nov
Grid connection at 6.6 kV	**Kansai T&D**	consultation drafted
Fuel bought in from neighbouring villages	**Uda City, others**	to approach
What students and researchers would measure	**NAIST (you)**	to discuss

ASK JAMES

What do we still not know about how the history actually went?

ROB 09 / 10 · WHERE WE STAND

Early planning. *No money raised, no contracts, no machines bought.*

NOW

Phase 0

Trust, sites, fuel talks, homework

DEC 2026

Phase 0 ends

Decision on first site and partners

2027

Prototype

50 kWe CHP and fuel trial, if site and partners settle

2028 →

Machines

A CTL trial if a fit is shown and financing is in place

LATER

Full scale

≈ 0.5–1.2 MWe and a larger compute load; no fixed date

> *¥0 committed. Dates are targets, not promises. The grant tracks we know of (METI GX regional, MoE zero-emission data center) are being studied, not applied for.*

← JAMES · 5

Goryōrin ran a hundred-year plan for 58 years. We are at year zero.

75 years to grow the forest.

25 years to grow the village.

Tell us *where we are wrong.*

Honest critique is more useful to us than polite interest. What would you measure? What would you doubt? Who else should we ask?

Rob Oudendijk · Mitsue-it · rob@mitsue.it · mitsue.it

All documents open: https://codeberg.org/YR-Design/mitsue-ai-data-center

← JAMES · 5

The last time anyone planned forestry on this horizon.